

Intro to Geoscience

Leah April



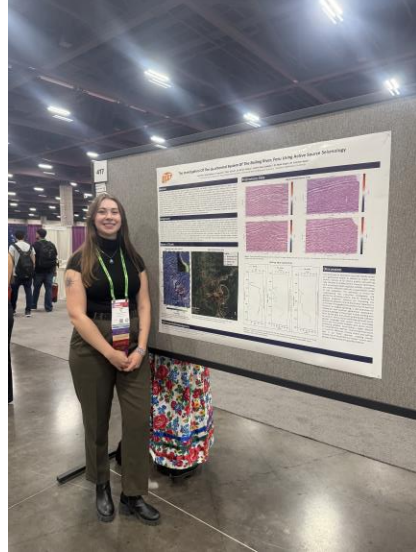
What is Geoscience?

- Also known as Earth sciences:
 - Atmosphere
 - Biosphere
 - Hydrosphere
 - Geosphere

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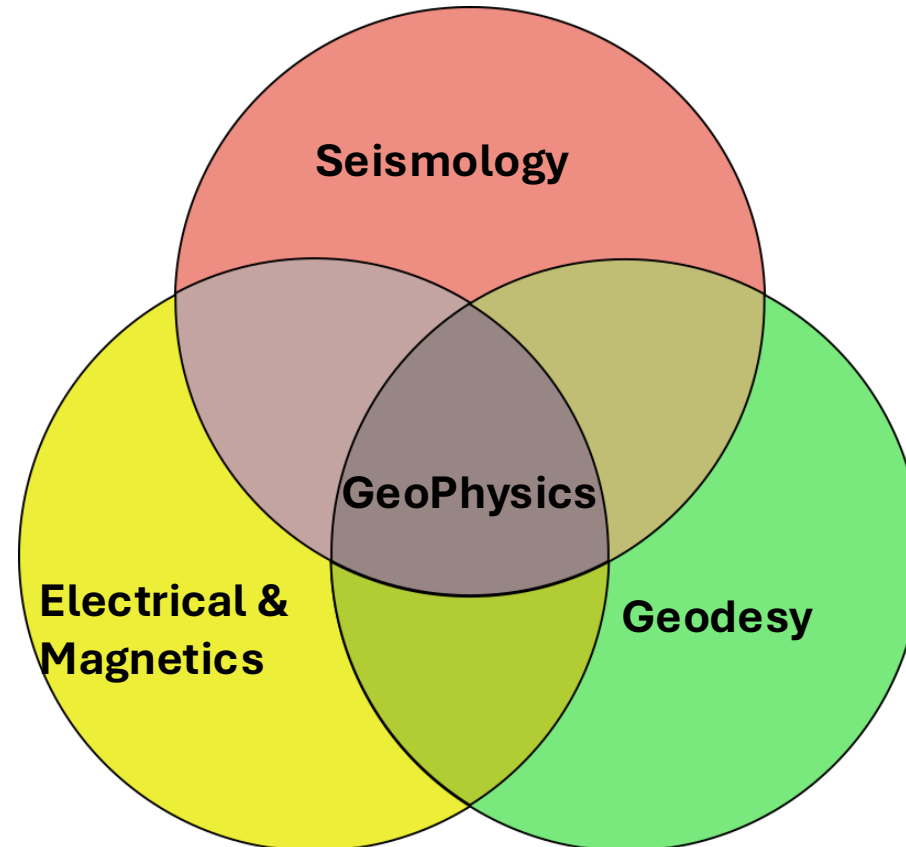


My Journey to Becoming a Geoscientist

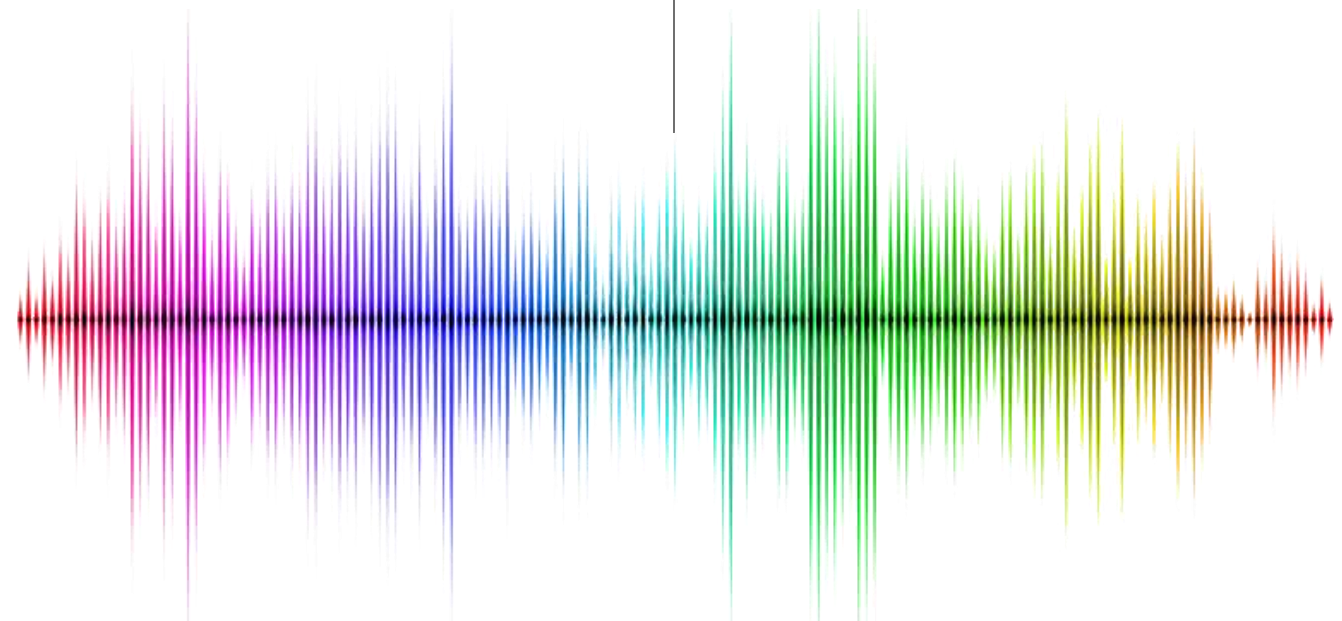
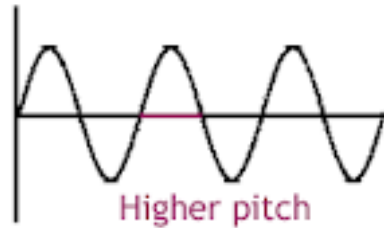
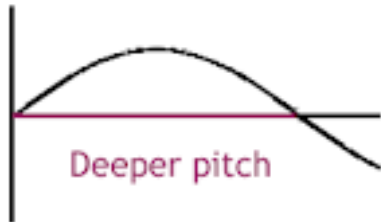
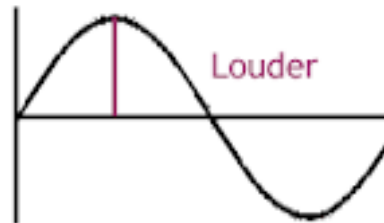
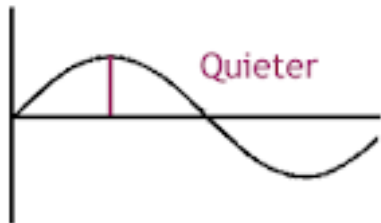
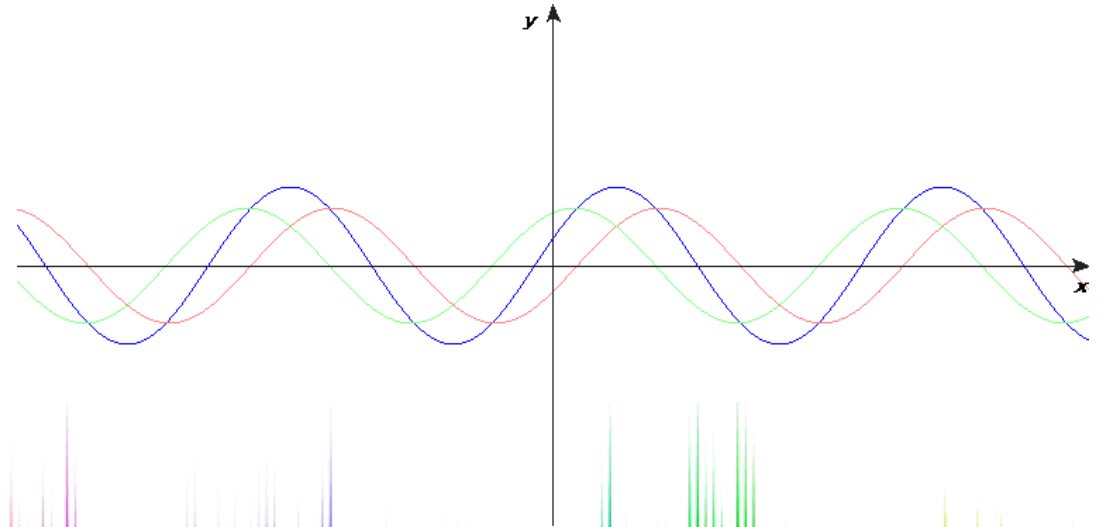
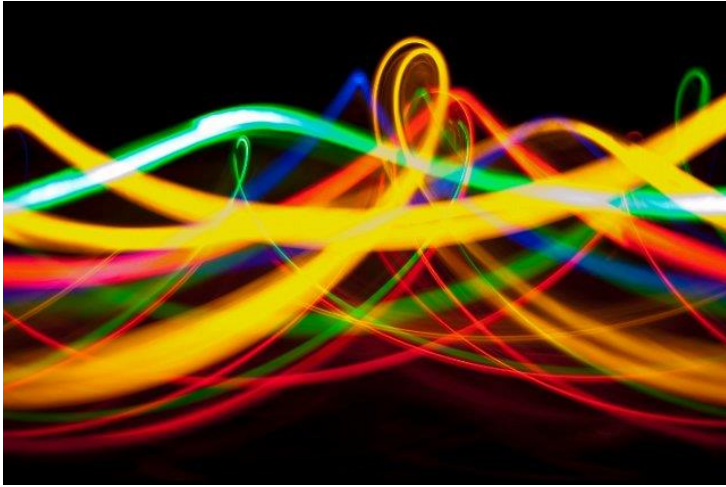


What is Seismology?

- **Seismology** is the study of earthquakes and seismic waves that move through and around the Earth.

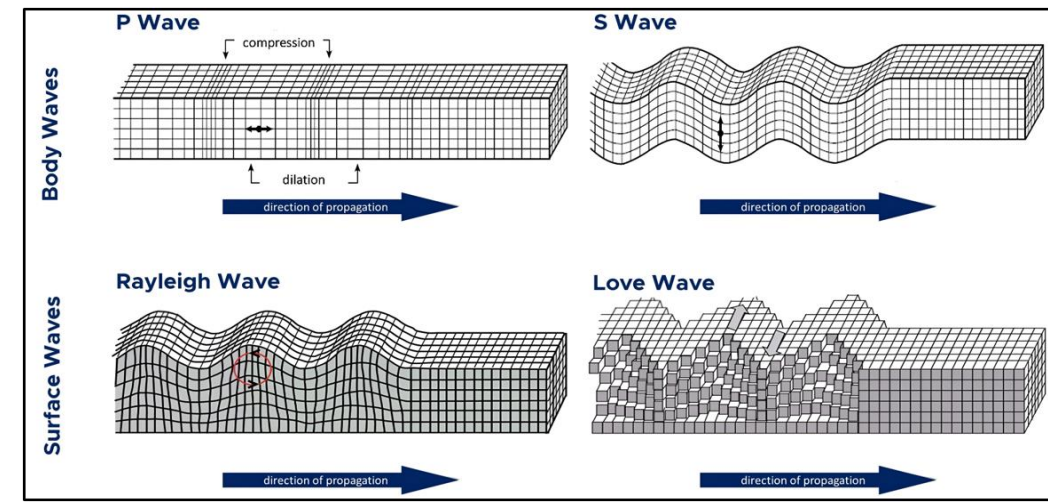
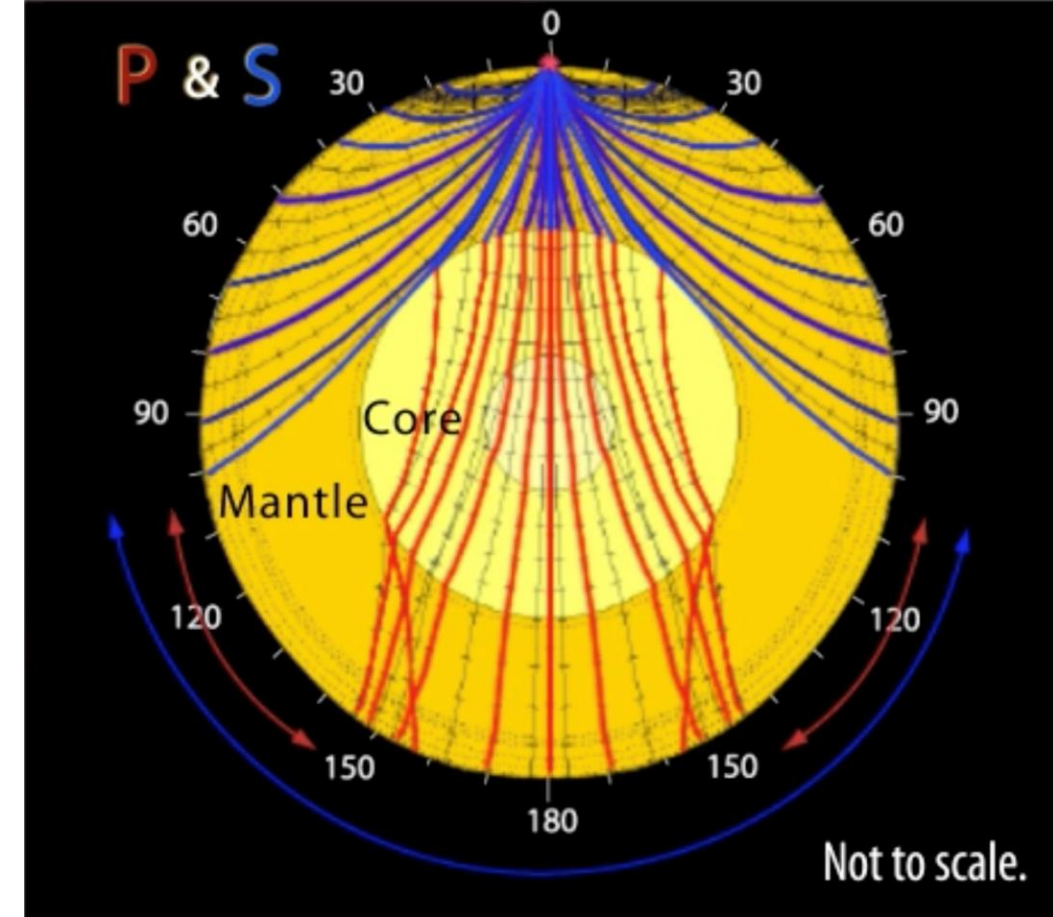


What are Some Types of Waves You Know?



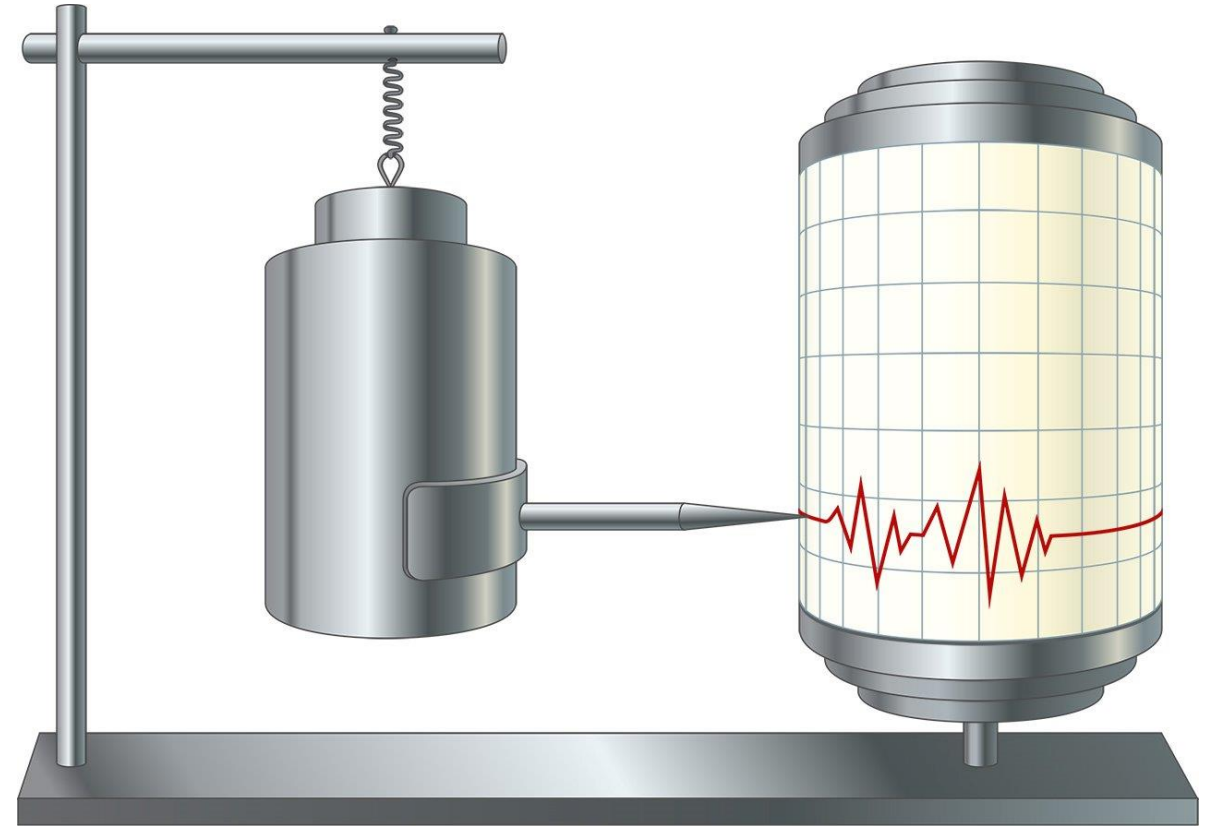
Seismology: How physics connects to geology

- **Basic concepts:** How waves move through different materials in the earth (or on mars and the moon).
 - Different waves and their properties:
- **Body Waves:**
 - **P-Waves (Primary Waves):** Compressional waves, fastest, travel through solids and liquids, classified as a longitudinal wave.
 - **S-Waves (Secondary Waves):** Shear waves, slower, travel only through solids, classified as a transverse wave.
- **Surface Waves:**
 - **Love Waves:** Horizontal shaking, cause damage to structures.
 - **Rayleigh Waves:** Rolling motion, cause both vertical and horizontal ground movement.



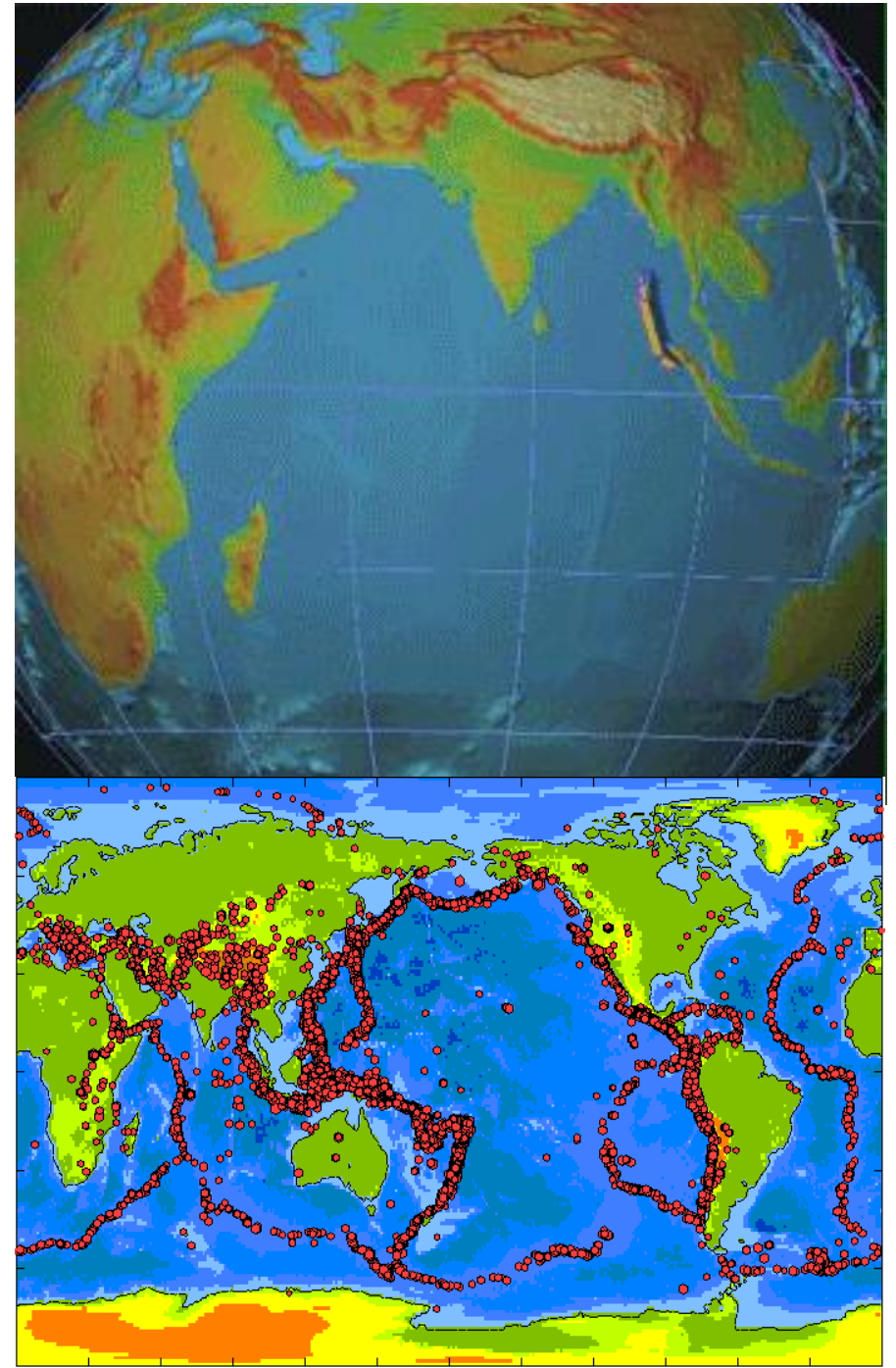
Measuring Earthquakes

- **How Do We Measure Earthquakes?**
- **Seismograph:** Instrument that records seismic waves.
- **Magnitude:** Measures the energy released (e.g., Richter scale, Moment Magnitude scale).
- **Intensity:** Measures the effects and damage caused (e.g., Modified Mercalli Intensity scale).



Let's Talk Earthquakes

- **What is an Earthquake, and where do they occur?**
- **Definition:** Sudden shaking of the ground caused by the movement of the Earth's tectonic plates.
- **Cause:** Release of energy stored in rocks due to stress build-up.



How do we really know what's inside the Earth?

- **Seismic Wave Propagation:**

- **P-Waves (Primary Waves):** Travel through solids and liquids; faster.
- **S-Waves (Secondary Waves):** Travel only through solids; slower.

- **Earth's Layers:**

- **Crust:** Solid rock.
- **Mantle:** Semi-solid rock.
- **Outer Core:** Liquid iron and nickel.
- **Inner Core:** Solid iron and nickel.

- **The velocities of the P and S waves change as they go through each distinctive earth layer.**

